

Solving Problems with Equations*Additional Practice 7A.01.05*

Name_____

Date_____

Period_____

For questions 1 through 3, solve using your preferred method, then check your solution by substitution.

1. $4(0.5p + 1.5) = 12$

2. $\frac{5}{2} = \frac{1}{4} (z - 8)$

3. $18 = \frac{3}{4} (36.4 - n)$

For questions 4 and 5, rewrite the expression by applying the distributive property.

4. $5(2y - 7)$

5. $(m + 3.5)(-6)$

Josephine paid \$650 to rent a beachfront cabana. For each guest who used the cabana, a \$19.40 resort fee was added to the bill. The final bill was \$882.80.

6. What was the total amount spent on the cabana?
7. Which cost depends on the number of guests?
8. Which cost is the same no matter how many guests use the cabana?
9. Write and solve an equation to find how many guests used the cabana. Use g for the number of guests.

Sade was asked to solve $\frac{2}{3}(b + 2.5) = 9.5$. Her work is shown. Use it for question 10.

$$\begin{aligned}\frac{2}{3}(b + 2.5) &= 9.5 \\ \frac{2}{3} \cdot b + \frac{2}{3} \cdot 2.5 &= 9.5 \cdot \frac{2}{3} \\ \frac{2}{3}b + \frac{5}{3} &= \frac{19}{3} \\ \frac{2}{3}b &= \frac{14}{3} \\ b &= 7\end{aligned}$$

10. Explain where Sade made her mistake, then find the correct solution.